

Elementary Problems

Exponent

Version 1.0

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Order

1. Find the orders of 3 modulo 7 and modulo 13 respectively.
2. Find the order of 5 modulo 64.
3. (IMO 1964 Problem 1)
 - (a) Find all positive integers n for which $2^n - 1$ is divisible by 7.
 - (b) Prove that there is no positive integer n for which $2^n + 1$ is divisible by 7.
4. (Kazakhstan MO 2010 Grade 10 Problem 4) It is given that for some $n \in \mathbb{N}$ there exists a natural number a , such that $a^{n-1} \equiv 1 \pmod{n}$ and that for any prime divisor p of $n - 1$ we have $a^{\frac{n-1}{p}} \not\equiv 1 \pmod{n}$.
Prove that n is a prime.
5. Let $x = \frac{m}{n}$ where m and n are relatively prime positive integers. Prove that the decimal representation of x has period at most $n - 1$.
6. Let p and q be primes numbers such that $p \mid 2^q - 1$. Prove that $p \equiv 1 \pmod{q}$.
7. Find all positive integers m and n such that $2^m - 3^n = 1$.
8. Find all positive integers m and n such that $3^m - 2^n = 1$.
9. Find all positive integers m and n such that $5^m - 12^n = 1$.
10. Find all positive integers m and n satisfying $3^m - 17^n = 64$.
11. Find all positive integers m and n such that $3^m + 22 = 5^n$.
12. (IMO shortlist 1991) Find all positive integer solutions x, y, z of the equation
$$3^x + 4^y = 5^z.$$
13. (Poland MO Finals 2014 Problem 5) Find all pairs (x, y) of positive integers that satisfy
$$2^x + 17 = y^4.$$
14. (IMO shortlist 1966) Find all pairs (x, y) of positive integers satisfying the equation
$$2^x = 3^y + 5.$$

15. (IMO 1977 Problem 3) Let n be a given integer > 2 , and let V_n be the set of integers $1 + kn$, where $k = 1, 2, \dots$. A number $m \in V_n$ is called indecomposable in V_n if there do not exist numbers $p, q \in V_n$ such that $pq = m$. Prove that there exists a number $r \in V_n$ that can be expressed as the product of elements indecomposable in V_n in more than one way. (Products which differ only in the order of their factors will be considered the same.)
16. (Korea Final Round 2005 Problem 5) Find all positive integers m and n such that both $3^m + 1$ and $3^n + 1$ are divisible by mn .

Lifting the Exponent Lemma

17. Find the order of 2 modulo 5^8 .
18. (IMO 1978 Problem 1) m and n are natural numbers with $1 \leq m < n$. In their decimal representations, the last three digits of 1978^m are equal, respectively, to the last three digits of 1978^n . Find m and n such that $m + n$ has its least value.
19. The integer $n = 33 \cdots 3$ consists of 3^{100} copies of 3's. Find the highest power of 3 that divides n .
20. (CGMO 2017 Problem 1)
 - (1) Find all positive integers n such that 4 divides $a^n - 1$ for any odd number a .
 - (2) Find all positive integers n such that 2^{2017} divides $a^n - 1$ for any odd number a .
21. Find all positive integers n such that $\frac{5^n + 1}{3^n}$ is an integer.
22. Find all positive integers m and n such that $3^{m+2} \mid 4^{3^n} - 1$ and $3^{n+2} \mid 4^{3^m} - 1$.
23. (IMO 1990 Problem 3) Determine all integers $n > 1$ such that

$$\frac{2^n + 1}{n^2}$$

is an integer.

Valuation

24. Find the largest positive integer n such that 2^n divides $72!$.
25. Find the largest positive integer n such that 5^n divides $1001^{200!} - 101^{200!}$.

26. Let k be a fixed positive integer and m a fixed nonnegative integer. Prove that there exists a positive integer $n \geq k$ such that the binomial coefficient $\binom{n}{k}$ is of the form $2^m a$ for some odd number a .
27. (Greece MO 2001 Problem 2) Prove that there are no positive integers a, b such that $(15a + b)(a + 15b)$ is a power of 3.
28. Find all positive integers m, n and k such that $2m^k = n^{k-1}$.
29. Find all positive integers a, b, c such that $abc^3 = a^3 + b^3$ and $(a, b, c) = 1$.

Record

v1.0: problems 1–29